

Chapter 4: Project Design

4.1 Overview of Project Design

This chapter focuses on decomposing a project into its various components and designing each part to visualize the complete structure and gain insights into how the project will appear when fully developed. The following diagrams and tools are utilized for this purpose:

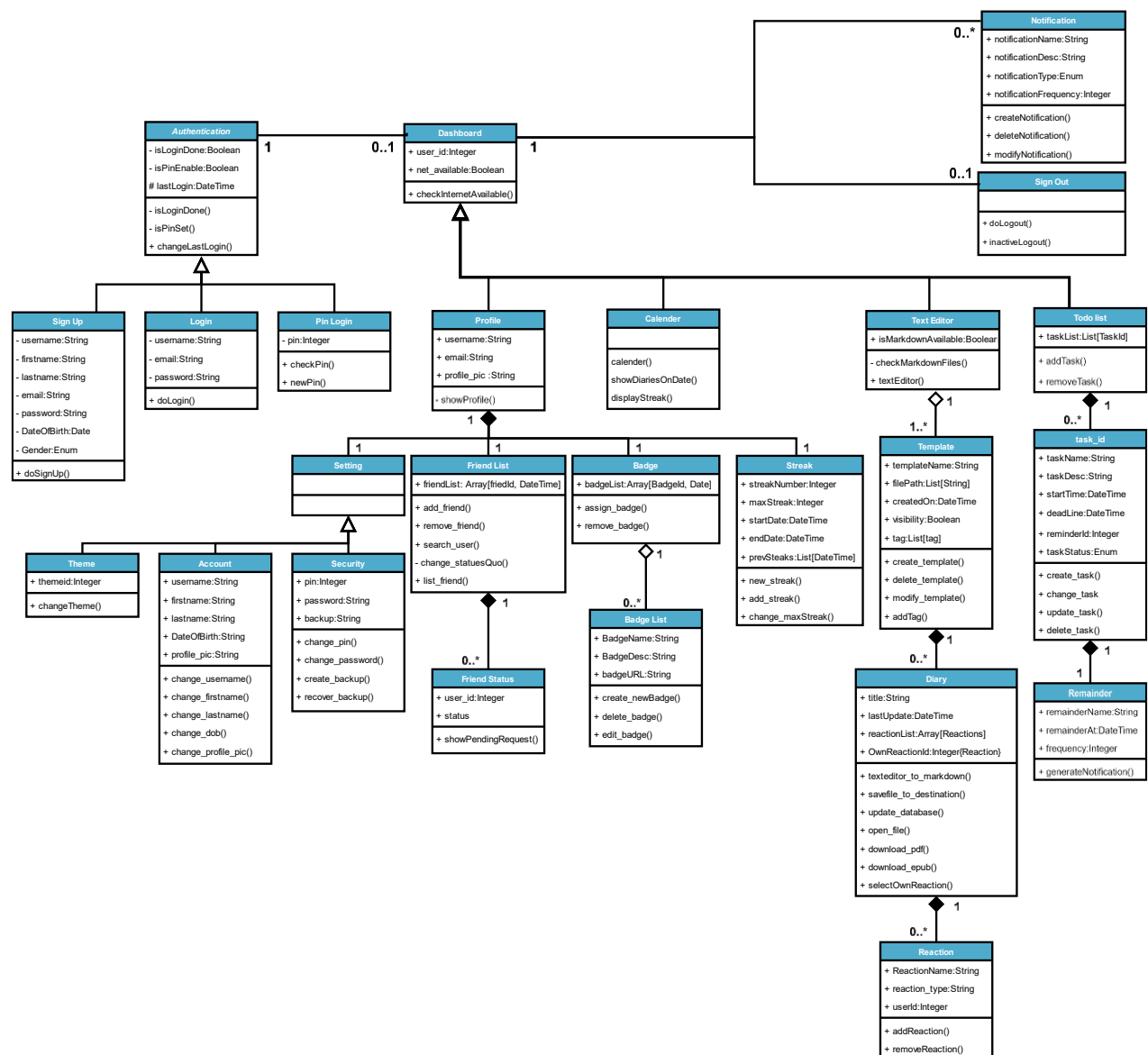
1. Class Diagram: Illustrates the classes, their attributes, methods, and the relationships between them.
2. State Diagram: Depicts the states of an object and the transitions between these states based on certain events.
3. Use Case Diagram: Shows the interactions between users (actors) and the system to achieve a goal.
4. Sequence Diagram: Visualizes the sequence of messages exchanged between objects in a particular scenario or process.
5. Activity Diagram: Represents the flow of control or activities in the system, typically used to describe the workflow of a process.
6. Data Dictionary: A detailed description of data elements, including their definitions, types, and relationships.
7. Data Flow Diagram (DFD):
 - a. Level 0: Provides a high-level overview of the system and its interactions with external entities.
 - b. Level 1: Breaks down the system into more detailed sub-processes, showing how data moves within the system.
8. Wireframes: Visual representations of the user interface (UI) layout, illustrating how the system will look from the user's perspective.

These tools are essential for understanding the system's architecture, interactions, and user interfaces, helping to build a clear vision of the final project.

4.2 Class Diagram

A UML class diagram is a visual tool that represents the structure of a system by showing its classes, attributes, methods, and the relationships between them. It helps everyone involved in a project—like developers and designers—understand how the system is organized and how its components interact.

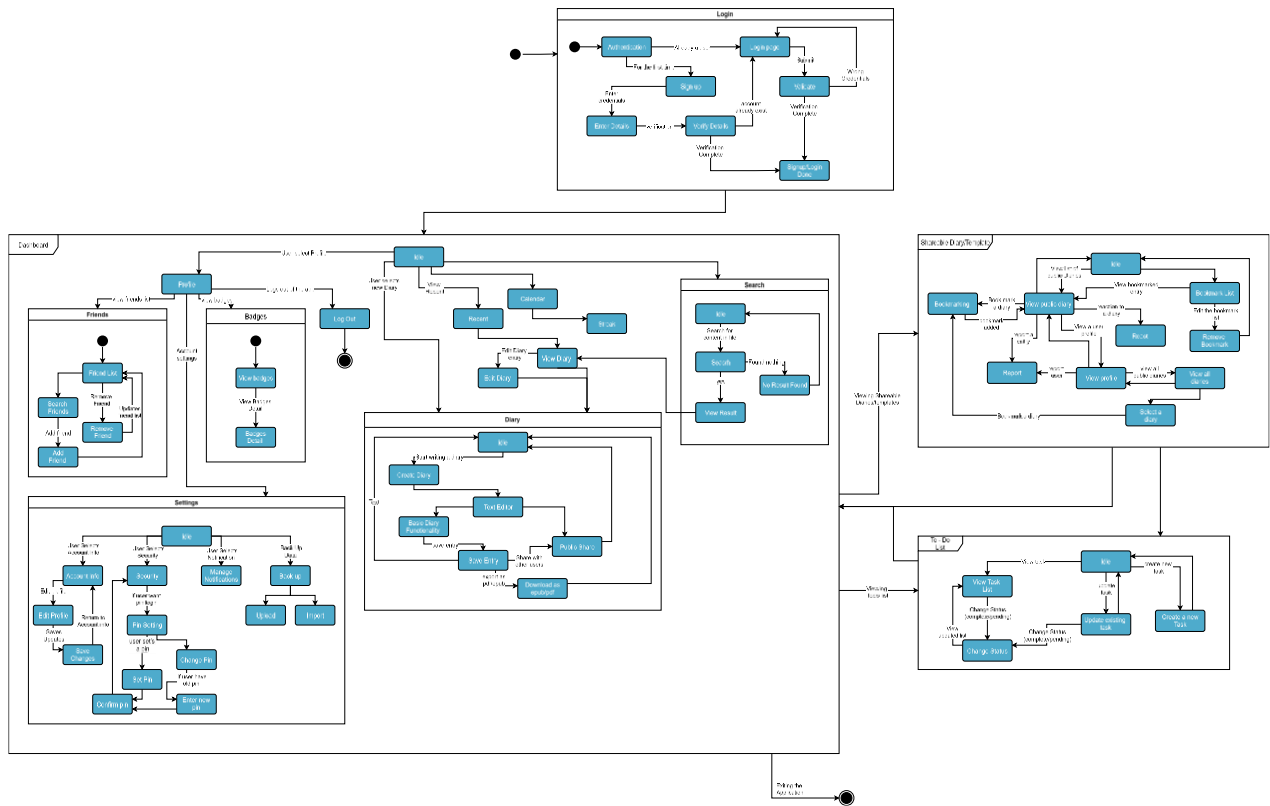
Below is our class Diagram.



4.2.1 Class Diagram

4.3 State Diagram

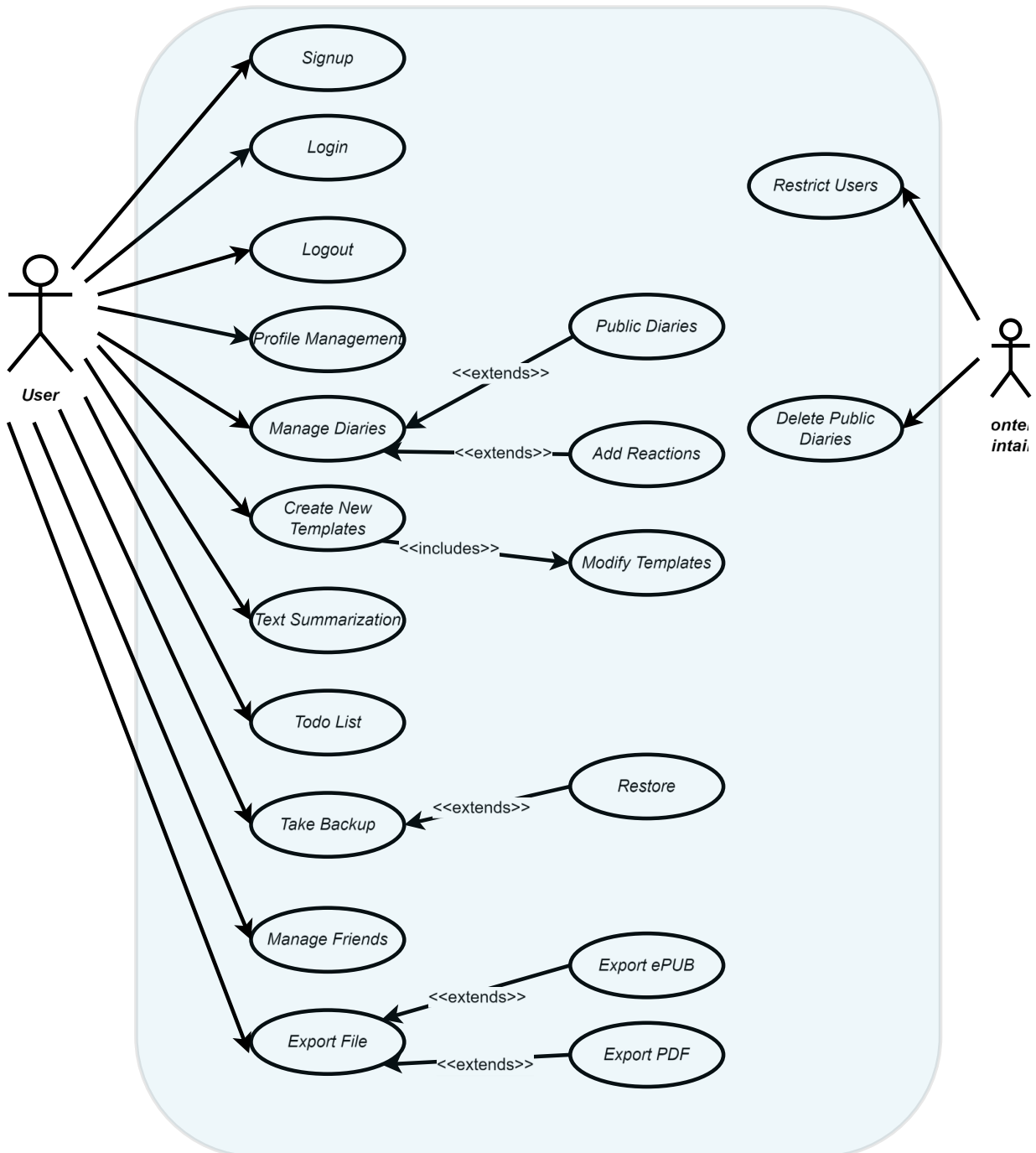
state diagram shows the behavior of classes in response to external stimuli.



4.3.1 State Diagram

4.4 Use Case Diagram

Use cases are a technique for capturing, modeling, and specifying the requirements of a system. A use case corresponds to a set of behaviors that the system may perform in interaction with its actors, and which produces an observable result that contributes to its goals.

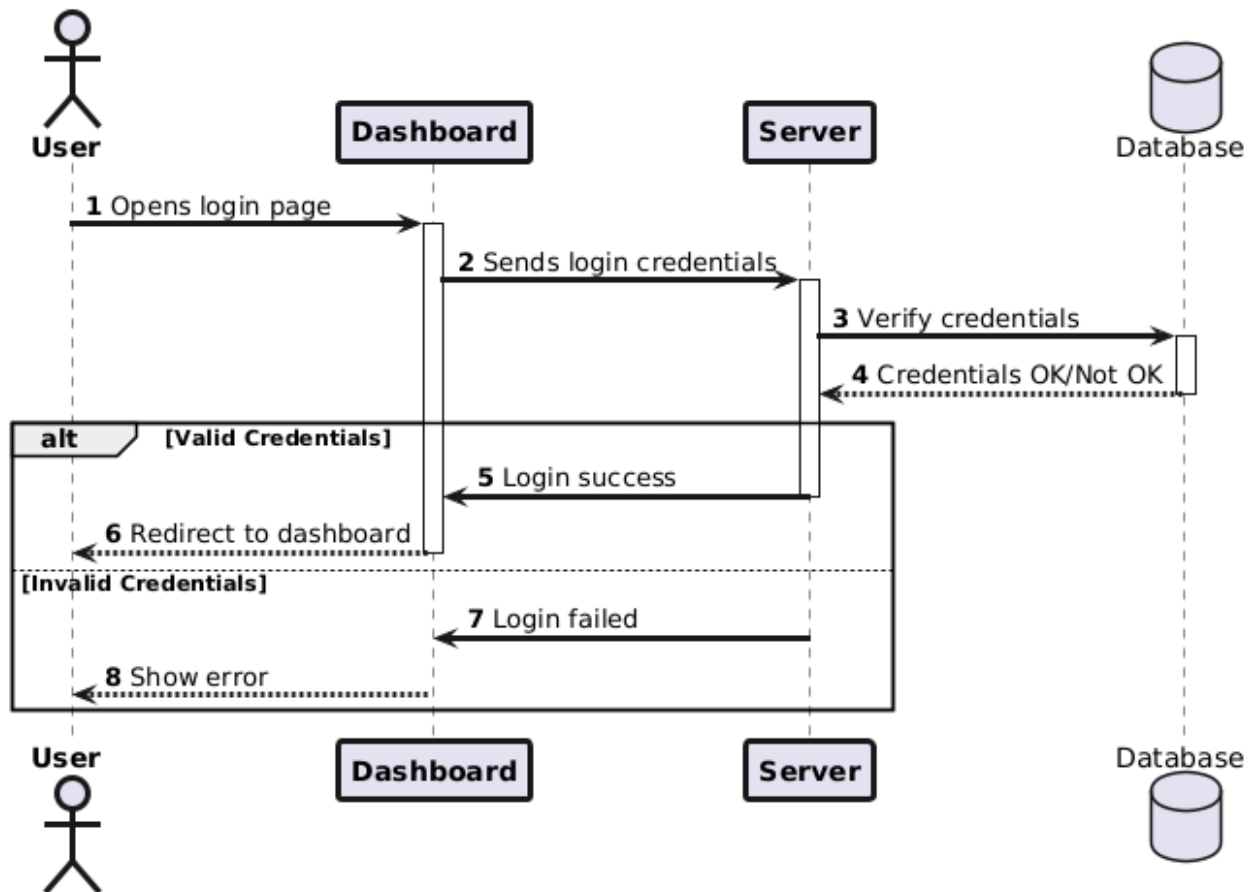


4.4.1 Use Case Diagram

4.5 Sequence Diagram

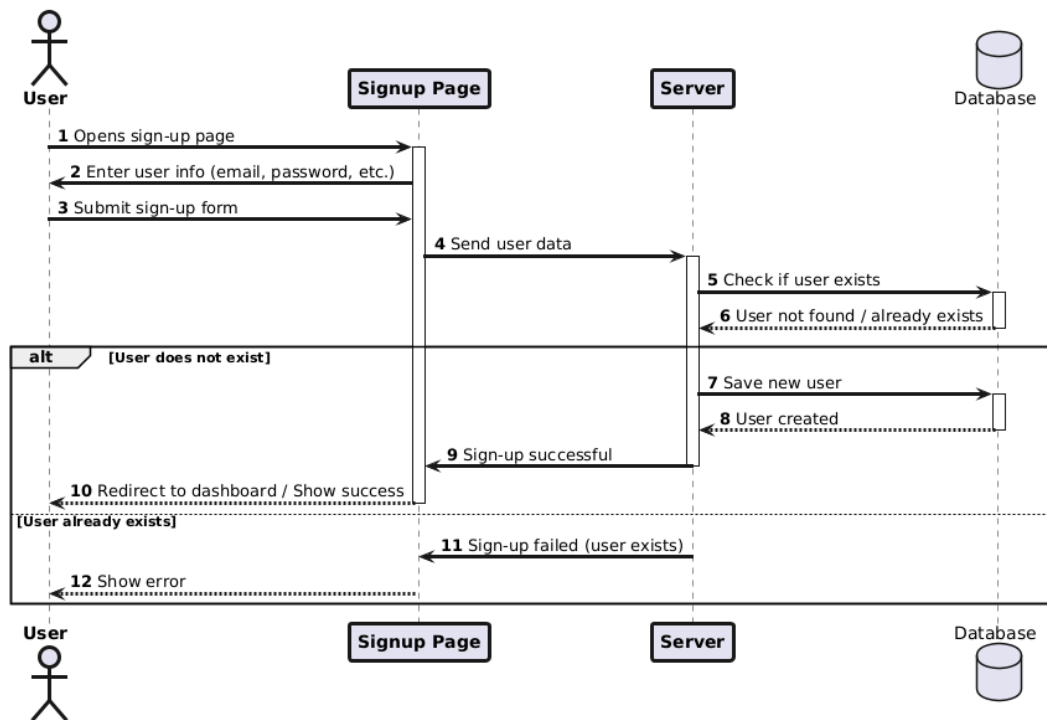
Sequence Diagram is an interaction diagram that details how operations are carried out -- what messages are sent and when.

i. Login Authentication



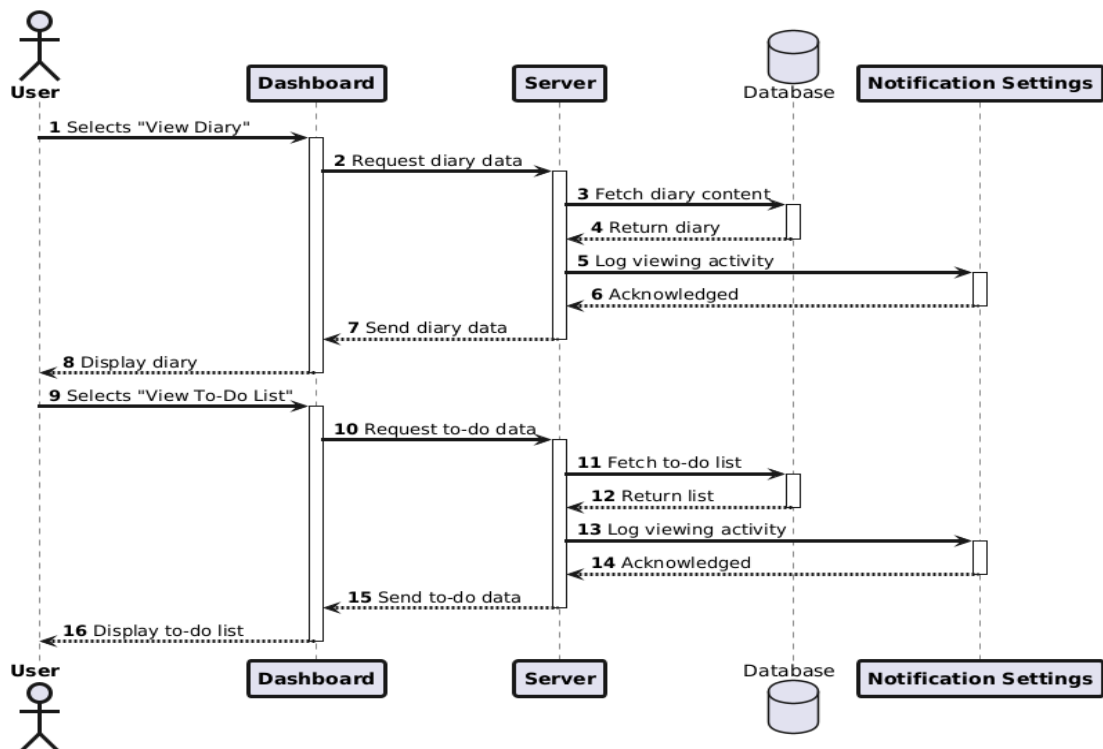
4.5.1 Sequence Diagram – Login Authentication

ii. Signup Authentication



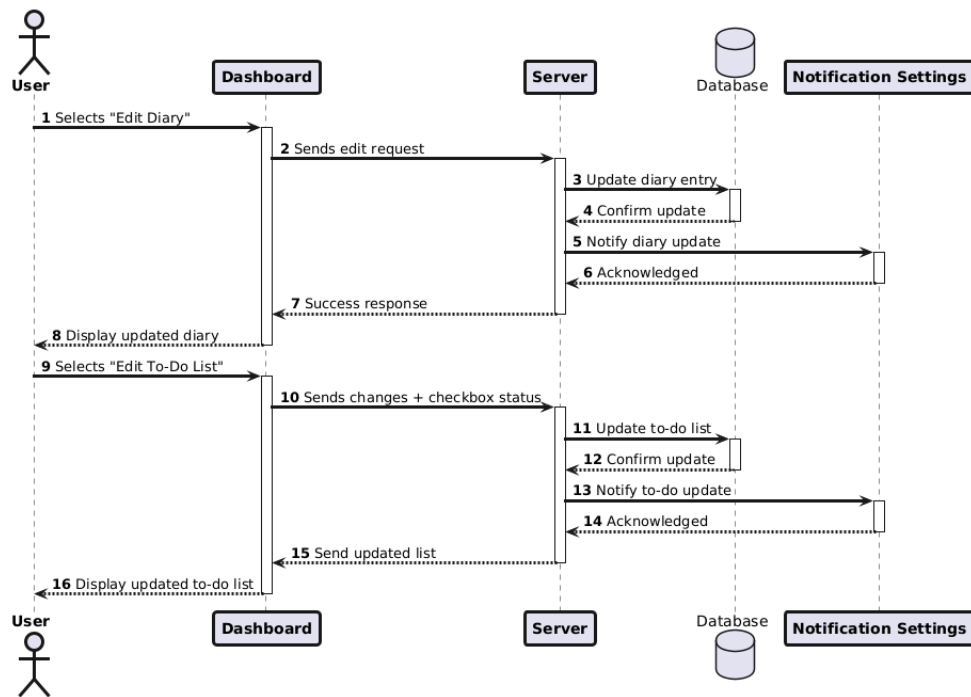
4.5.2 Sequence Diagram – Signup Authentication

iii. View Diary and View Todo List



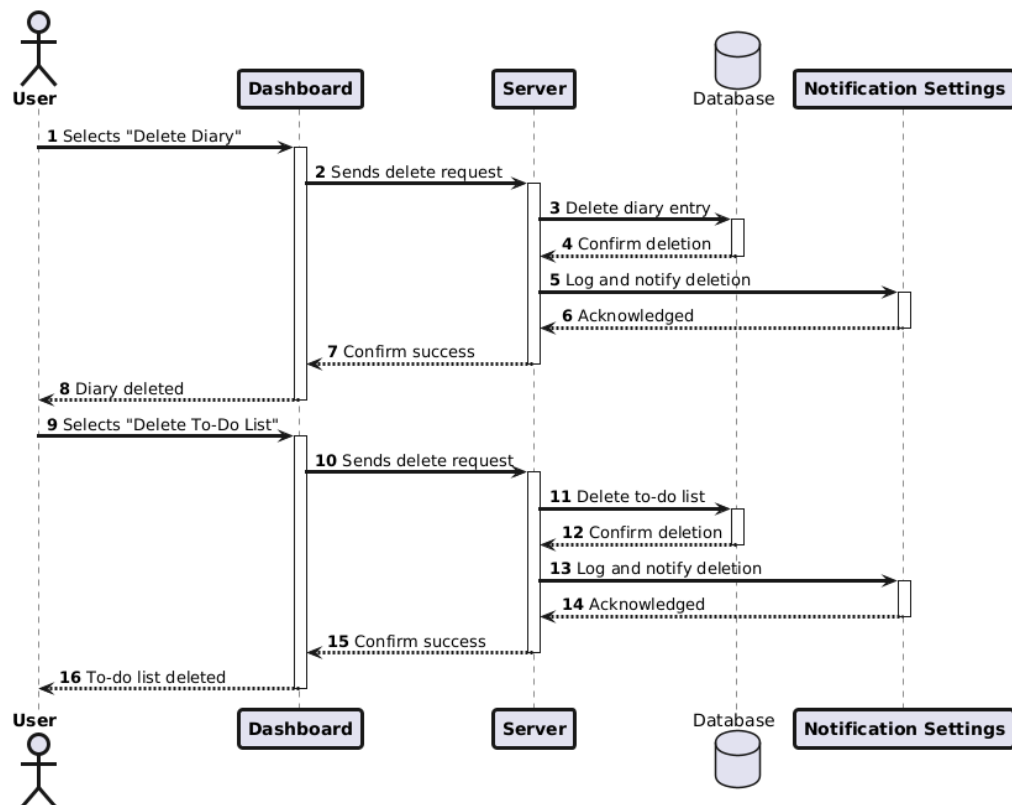
4.5.3 Sequence Diagram – View Diary and ToDo

iv. Edit Diary and Edit Todo List



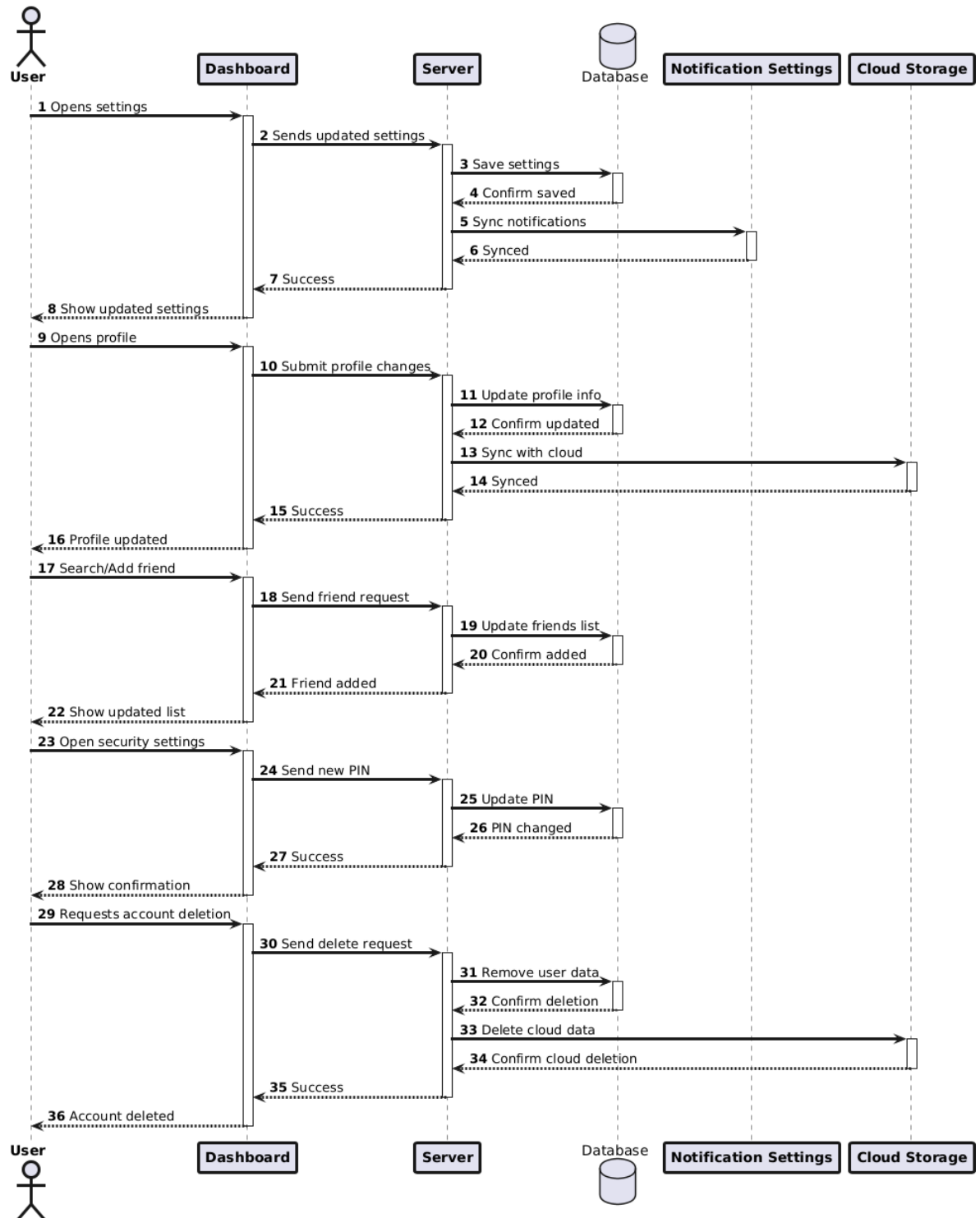
4.5.4 Sequence Diagram – Edit Diary and Edit ToDo

v. Delete Diary and Delete Todo List



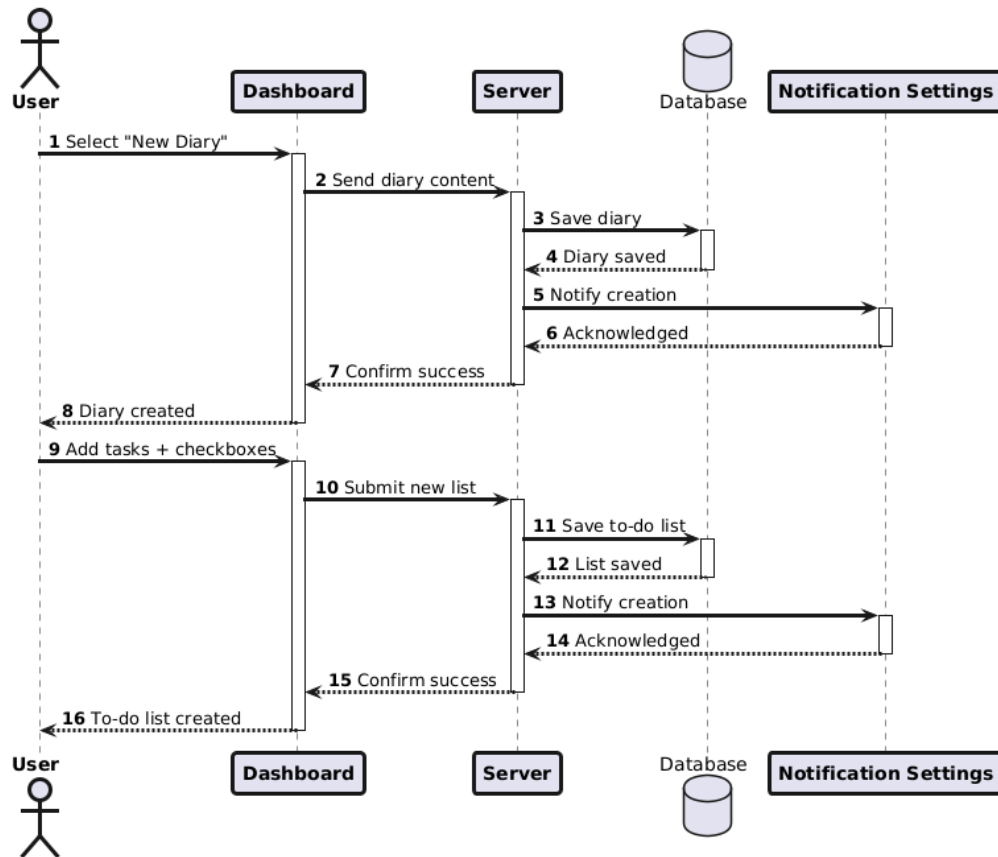
4.5.5 Sequence Diagram – Delete Diary and ToDo

vi. Edit Settings, Profile, Add Friends, and Account Deletion



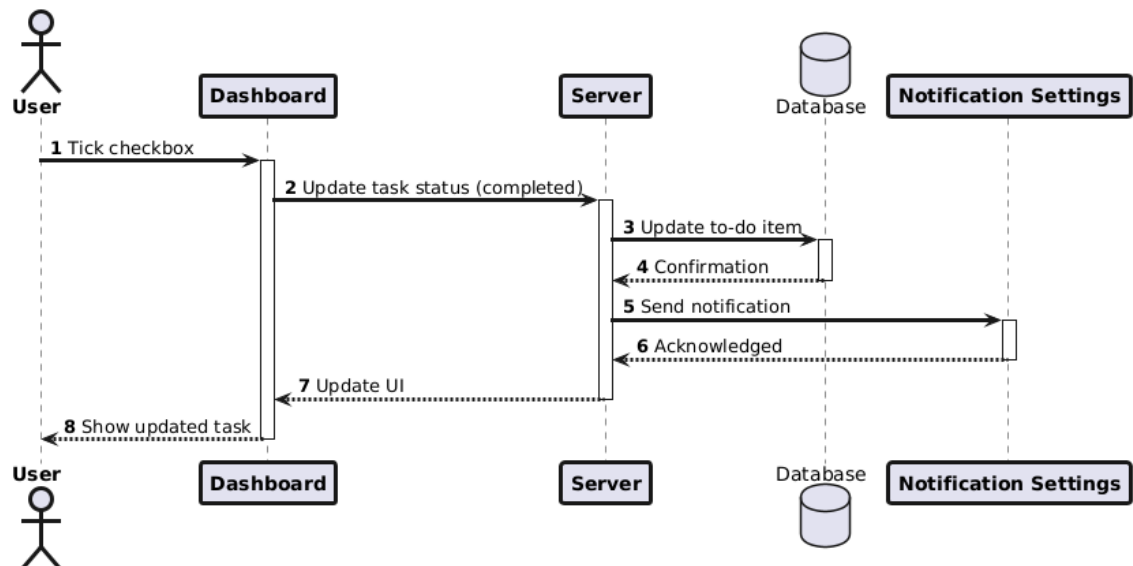
4.5.6 Sequence Diagram - Edit Settings, Profile, Add Friends, and Account Deletion

vii. Create Diary and ToDo



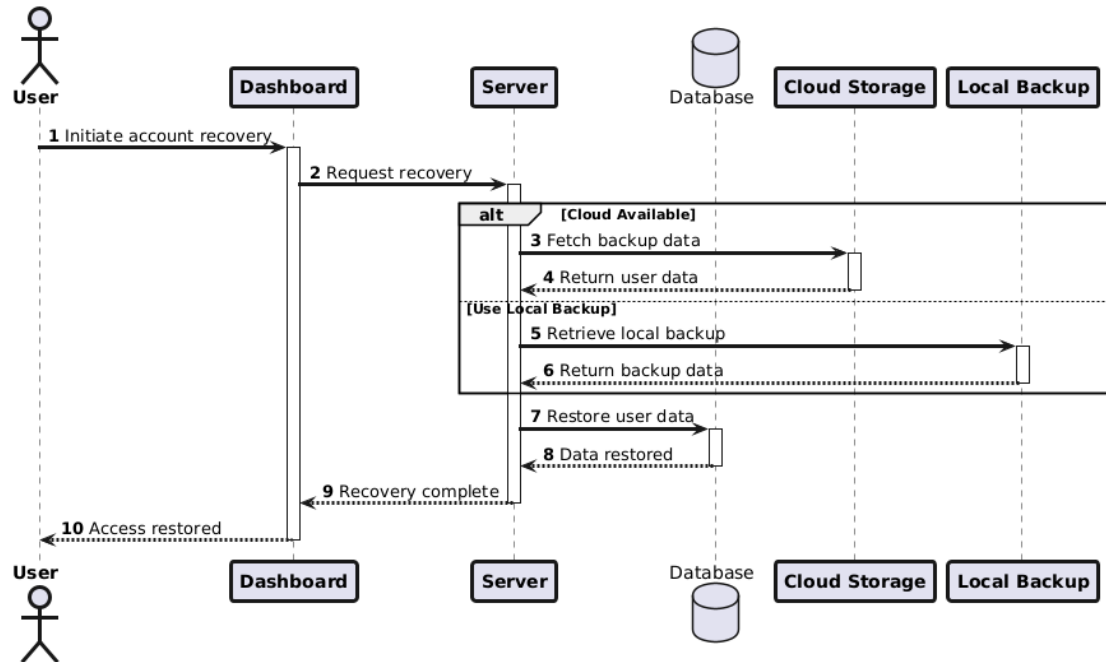
4.5.7 Sequence Diagram – Create Diary and ToDo

viii. Mark ToDo as Complete



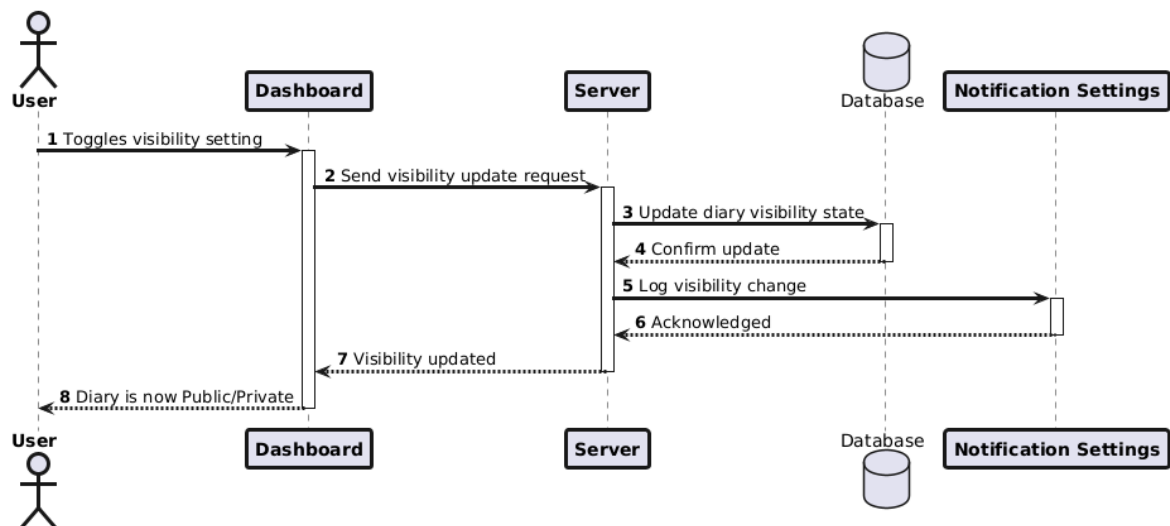
4.5.8 Sequence Diagram – Mark ToDo Complete

ix. Account Recovery



4.5.9 Sequence Diagram – Account Recovery

x. Change Diary Visibility



4.5.10 Sequence Diagram – Public Diary

4.7 Data Dictionary

A Data Dictionary is a collection of names, definitions, and attributes about data elements that are being used or captured in a database, information system, or part of a research project.

We have total of 15 data dictionaries needed in our project to store data and implement the smooth transitions in the data flow.

i. Account

<u>Account</u>				
Element	Data Type	Key	Auto Increment	Description
user_id	integer(8)	PK	Yes	A unique identifier for each user
username	varchar(32)			The unique name chosen by the user to be identified by in the app.
user_email	varchar(64)			The user's email address
user_firstname	varchar(32)			The first name of the user.
user_lastname	varchar(32)			The last name of the user.
user_gender	enum			The gender of the user.
user_dob	DATE			The date of birth of the user.
user_pfp_url	varchar(255)			The URL of the user's profile picture, used to display the profile image.
join_date	timestamp			The date and time when the user created their account.
theme_id	varchar(4)			A reference to the user's chosen theme.

Table 4.7.1: Account Table

ii. Login

<u>Login</u>				
Element	Data Type	Key	Auto Increment	Description
user_id	integer(8)	FK		A unique identifier for each user
username	varchar(32)			The unique name chosen by the user to be identified by in the app.
password	varchar(64)			The user's password for account access.
user_pin	varchar(4)			The PIN set by the user for an additional login method.
pin_enabled	boolean			Indicates whether the PIN login option is enabled (true/false).
login_available	boolean			Specifies if the user is currently allowed to log in.
login_attempy	integer(1)			Tracks the number of failed login attempts(max 3).

Table 4.7.2: Login Table

iii. Diary

<u>Diary</u>				
Element	Data Type	Key	Auto Increment	Description
diary_id	integer(8)	PK	Yes	A unique identifier for each diary entry.
user_id	integer(8)	FK		A unique identifier for each user.
profile_visibility	boolean			Indicates whether the diary entry is public or private.
created_at	timestamp			The date and time when the diary entry was created.
last_modified	timestamp			The date and time when the diary entry was last modified.
content_link	varchar(255)			A link to the stored content of the diary.

Table 4.7.3: Diary Table

iv. Tag

<u>Tag</u>				
Element	Data Type	Key	Auto Increment	Description
tag_id	integer(8)	PK	Yes	A unique identifier for each tag.
tag_name	varchar(24)			The name of the tag used to label diary entries.
tag_description	varchar(64)			A brief description of the tag's purpose or meaning.

Table 4.7.4: Tag Table

v. Tag Entries

<u>Tag Entries</u>				
Element	Data Type	Key	Auto Increment	Description
diary_id	integer(8)	FK		A unique identifier for the diary entry associated with the tag.
tag_id	integer(8)	FK		A unique identifier for the tag.

Table 4.7.5: Tag Entries

vi. Notification

<u>Notification</u>				
Element	Data Type	Key	Auto Increment	Description
noti_id	integer(8)	PK	Yes	A unique identifier for each notification.
noti_type	varchar(24)			The type of notification(reminder, alert).
noti_desc	varchar(100)			A description of the notification.
noti_time	date			The time when the notification is scheduled.
noti_date	date			The date when the notification is scheduled.
noti_frequency	integer(2)			The frequency of the notification.

Table 4.7.6: Notification Table

vii. Badge

<u>Badge</u>				
Element	Data Type	Key	Auto Increment	Description
badge_number	integer(8)	PK	Yes	A unique identifier for each badge.
badge_name	varchar(24)			The name of the badge earned by the user.
Badge_type	varchar(24)			The type or category of the badge.
Badge_link	varchar(255)			A link to the image or content associated with the badge.

Table 4.7.7 Badge Table

viii. User Badge Info

<u><i>User Badge Info</i></u>				
Element	Data Type	Key	Auto Increment	Description
badge_id	integer(8)	PK	Yes	A unique identifier for each badge info record.
badge_number	integer(8)	FK		A unique identifier for the badge associated with the user.
user_id	integer(8)	Fk		A unique identifier for each user.
received_date	timestamp			The date and time when the badge was received.

Table 4.7.8: User Badge Info Table

ix. Backup

<u>Backup</u>				
Element	Data Type	Key	Auto Increment	Description
backup_id	integer(8)	PK	Yes	A unique identifier for each backup record.
user_id	integer(8)	FK		A unique identifier for each user.
backup_date	timestamp			The date and time when the backup was created.
backup_location	varchar(255)			The location where the backup is stored.
backup_status	boolean			Indicates whether the backup(successful or failed).

Table 4.7.9: Backup Table

x. Friend List

<u>Friend List</u>				
Element	Data Type	Key	Auto Increment	Description
friend_number	integer(4)	PK	Yes	A unique identifier for each friend record.
user_id	integer(8)	FK		A unique identifier for each user.
friend_userid	enum			
request_status	boolean			Indicates whether the friend request is pending, accepted, or rejected.

Table 4.7.10: Friend Table

xi. Theme

<u>Theme</u>				
Element	Data Type	Key	Auto Increment	Description
theme_id	integer(8)	PK	Yes	A unique identifier for each theme.
theme_name	varchar(24)			The name of the theme.
theme_desc	varchar(64)			A description of the theme.
format_link	varchar(255)			A link to the format or design file associated with the theme.

Table 4.7.11: Theme Table

xii. Template

<u>Template</u>				
Element	Data Type	Key	Auto Increment	Description
template_id	integer(8)	PK	Yes	A unique identifier for each template.
user_id	integer(8)	FK		A unique identifier for each user.
template_name	varchar(24)			The name of the template.
template_link	varchar(255)			A link to the template content or file.

Table 4.7.12: Template Table

xiii. Reaction

<u><i>Reaction</i></u>				
Element	Data Type	Key	Auto Increment	Description
react_number	integer(8)	PK	Yes	A unique identifier for each reaction record.
react_id	integer(8)	FK		A unique identifier for the reaction type.
user_id	integer(8)	FK		A unique identifier for each user.
dairy_id	integer(8)	FK		A unique identifier for each diary entry.

Table 4.7.13: Reaction Table

xiv. Reaction Type

<u><i>Reaction Type</i></u>				
Element	Data Type	Key	Auto Increment	Description
react_id	integer(8)	PK	Yes	A unique identifier for each reaction type.
react_name	varchar(24)			The name of the reaction.
react_desc	varchar(64)			A description of the reaction.

Table 4.7.14: Reaction Type Table

xv. **Streak**

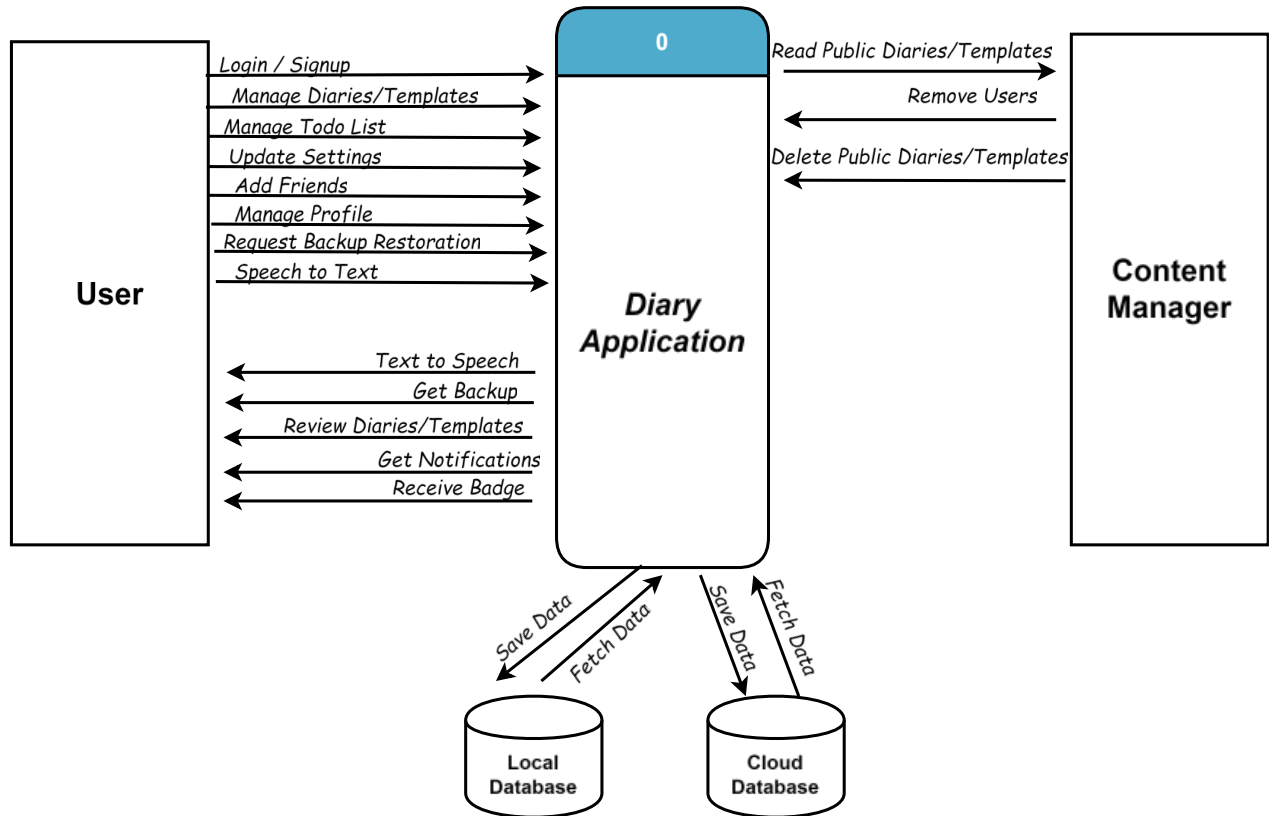
<u><i>Streak</i></u>				
Element	Data Type	Key	Auto Increment	Description
Streak ID	Integer(8)	PK	Yes	A unique identifier for each Streak.
user_id	integer(8)			
Date Started	DateTime			

Table 4.7.15: Streak Table

4.8 Data Flow Diagram

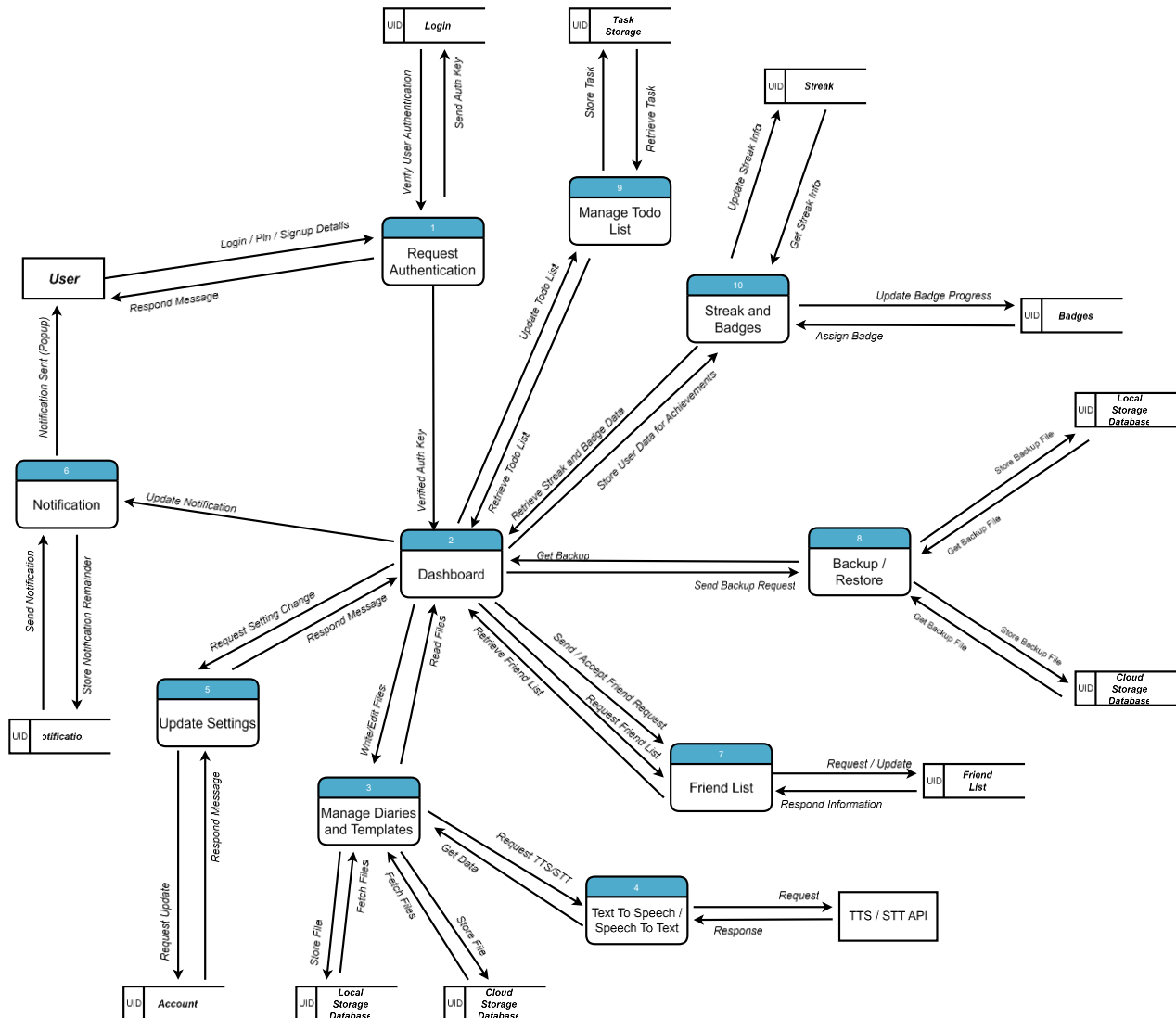
Data Flow Diagram (DFD) represents the flow of data within information systems.

4.8.1 Level 0 DFD



4.8.1 Level 0 DFD

4.8.2 Level 1 DFD

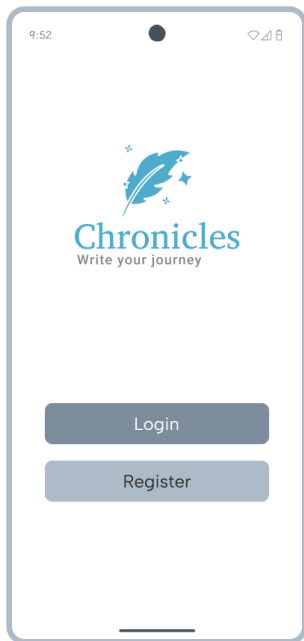


4.8.2 Level 1 DFD

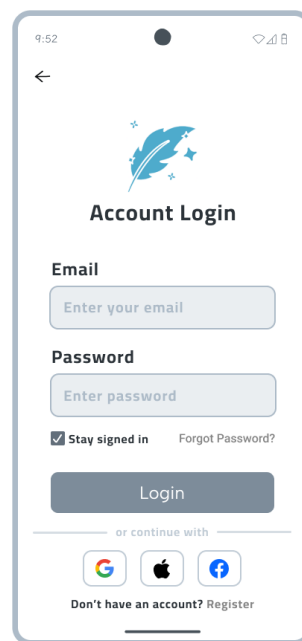
4.9 Wireframes

A wireframe is a rough schematic created in the early stages of digital product design to help visualize and communicate the structure of a product or website.

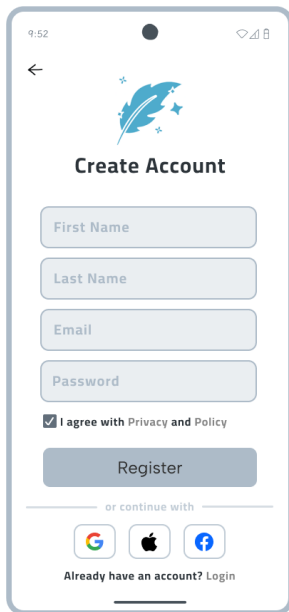
i. Login and Register Wireframes



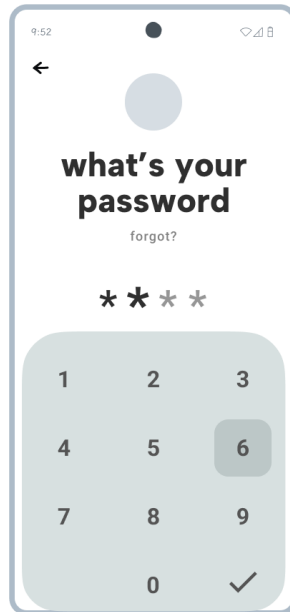
4.9.1 Welcome Screen



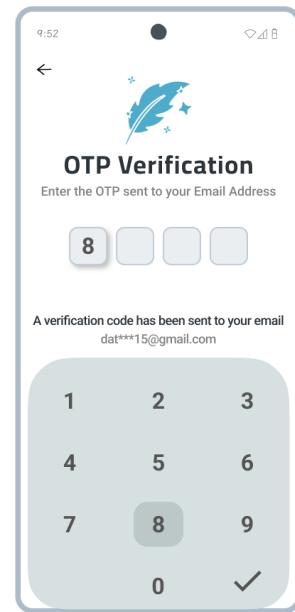
4.9.2 Login Screen



4.9.3 Register Screen

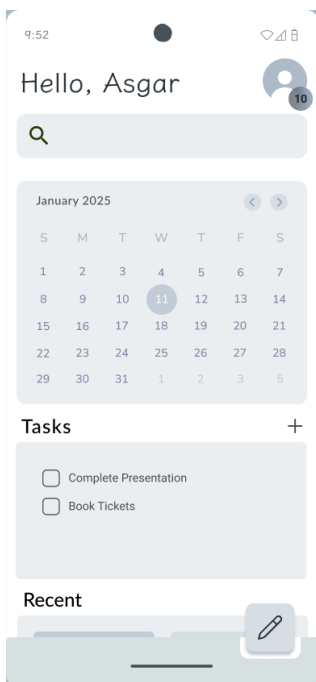


4.9.4 Pin Screen

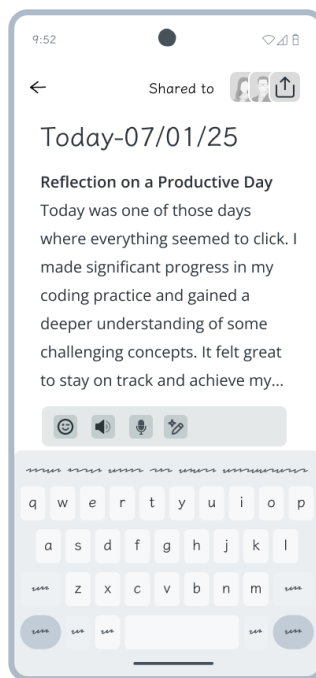


4.9.5 OTP Verification Screen

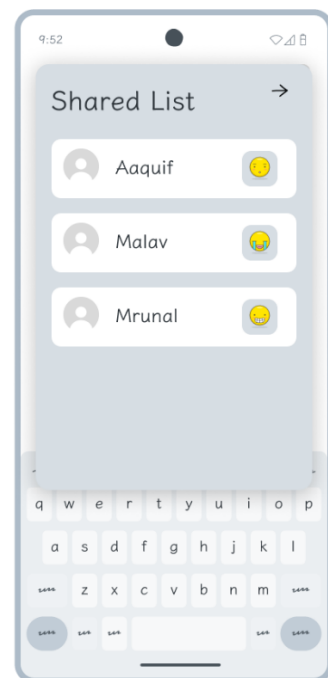
ii. Dashboard, Diary Writing, Sharing Diary, Settings Wireframe



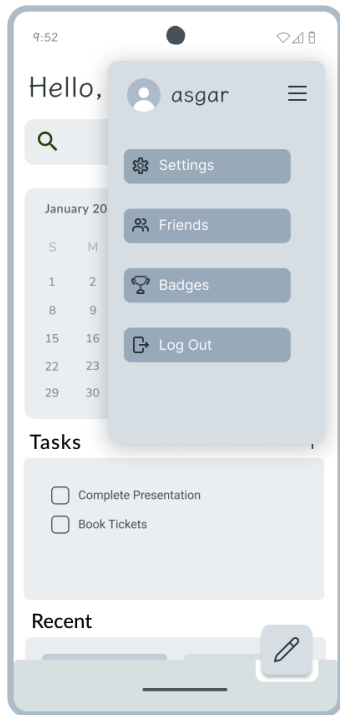
4.9.6 Dashboard Screen



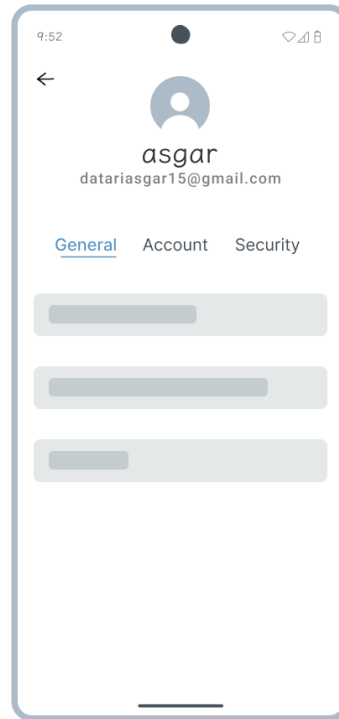
4.9.7 Text Editor Screen



4.9.8 Share With Friends In Text Editor Screen

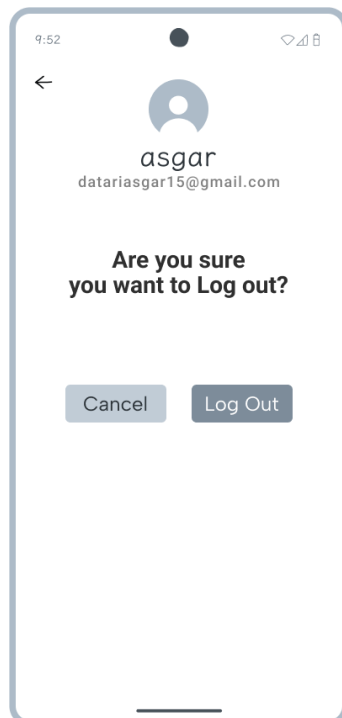


4.9.9 Settings Popup Screen



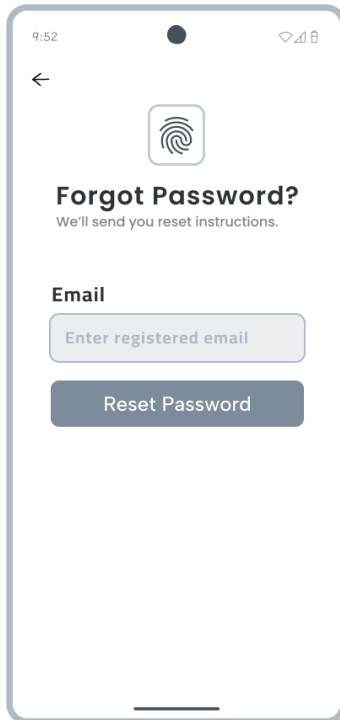
4.9.10 Profile Settings Screen

iii. Logout wireframe

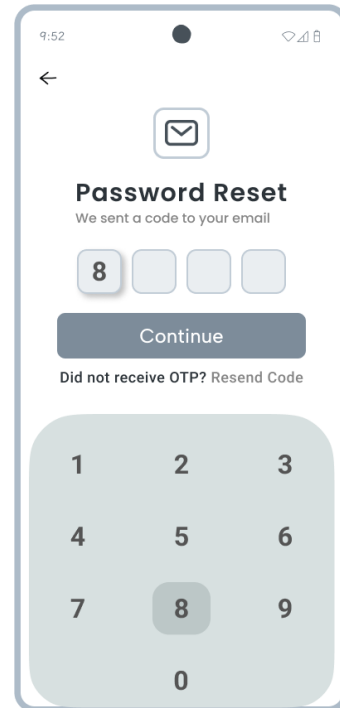


4.9.11 Logout Screen

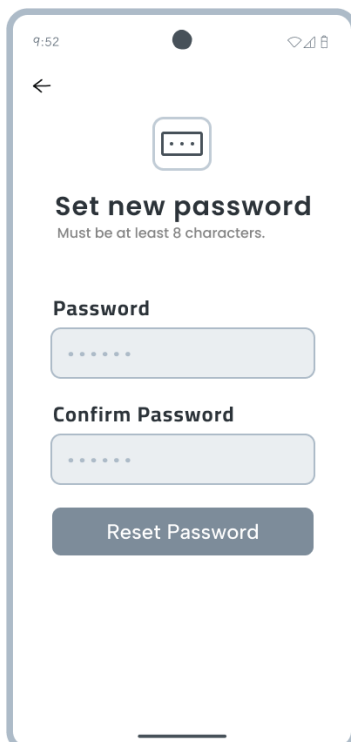
iv. Change Password Wireframe



4.9.12 Forgot Password Screen

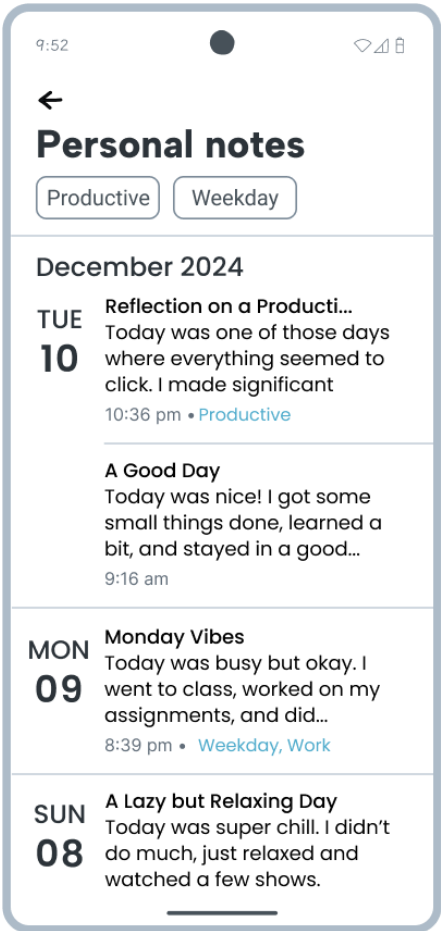


4.9.13 Password Reset Verification Screen



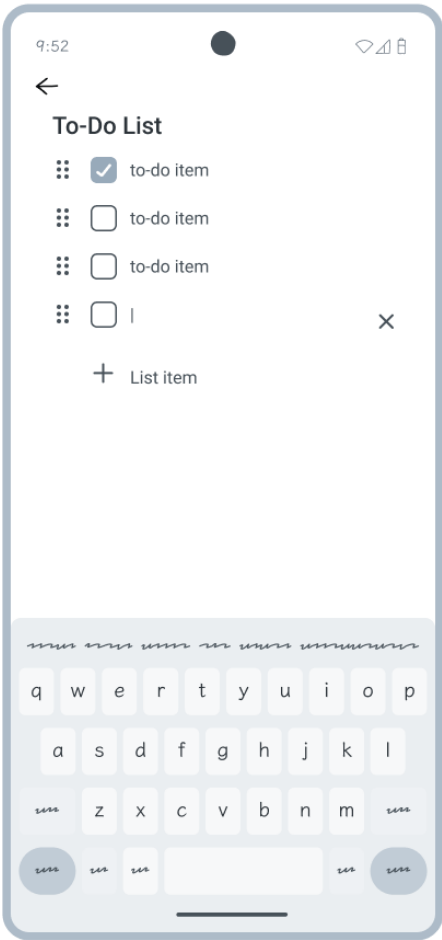
4.9.14 Change Password Screen

v. Personal Notes Wireframe

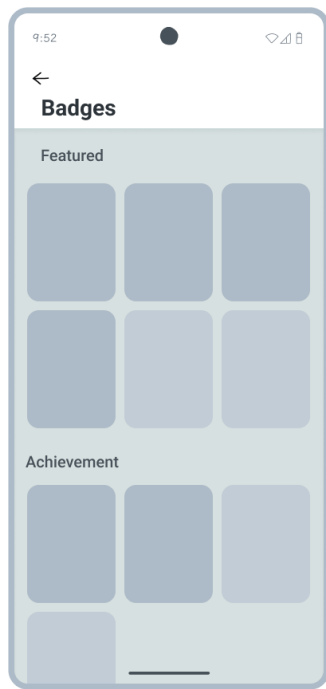


4.9.15 Archive Screen

vi. ToDo List Wireframe



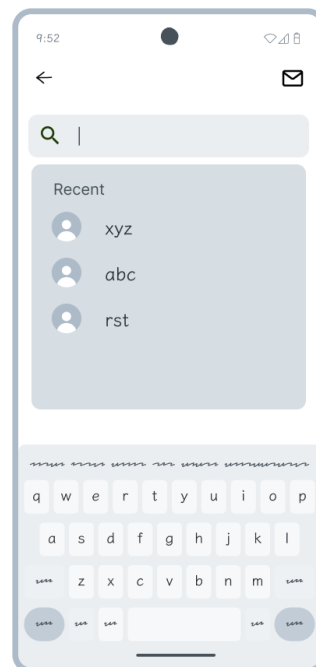
4.9.16 ToDo List

vii. Badges Wireframe

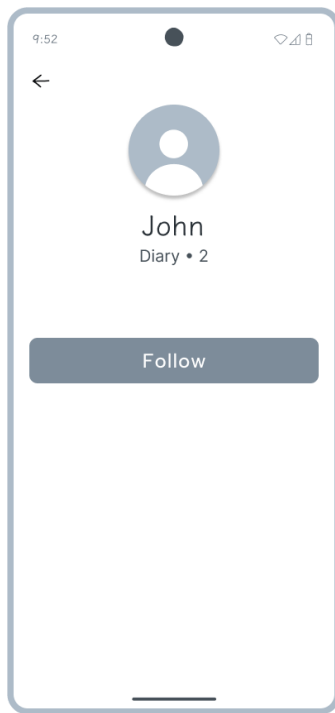
4.9.17 Badge Screen

viii. Search Friends

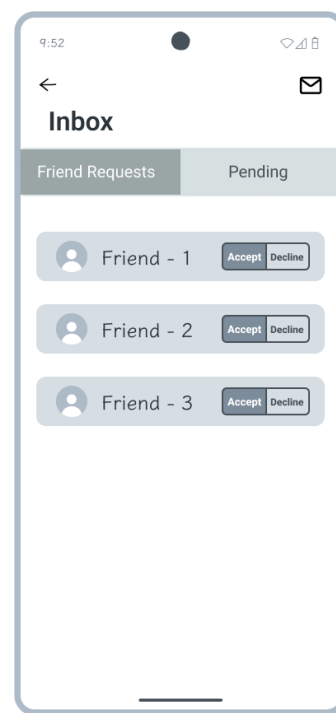
4.9.18 Friend List Screen



4.9.19 Search Friend Screen



4.9.20 Follow Friend Screen



4.9.21 Accept, Pending Friend Inbox Screen

Wireframes were our design layouts, we used to make better UIUX for our application. We changed aspects in wireframes based on user friendliness, and simple to keep UI. Our Project Screenshots are attached in Chapter 8.